

Priyanshu Pramanik

Software Engineer

Kolkata, India | +91 6204058150 | priyanshuallex@gmail.com
[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Detail-oriented Software Engineer specializing in full-stack web development, relational database design (PostgreSQL, MySQL), and intelligent automation. Proven capability building high-performance React interfaces, scalable Node.js/Express API backends, and Python ML microservices. Strong academic record (8.95 CGPA) plus technical internship experience designing data pipelines and automating workflows.

CORE TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, SQL (PostgreSQL, MySQL), HTML5, CSS3
Frontend: React, React 19, Tailwind CSS, DOM Manipulation, Responsive Web Design
Backend & APIs: Node.js, Express.js, FastAPI, RESTful APIs, JWT Authentication, MVC Architecture
Databases: PostgreSQL, Neon DB, MySQL (indexing, schema design, query optimization), MongoDB
Tools & Cloud: Git, GitHub, Docker, Postman, Vercel, Render, CI/CD pipelines
Systems & Auto: ETL Pipelines, Data Validation (Pandas, NumPy), SMTP Outreach Automation

PROFESSIONAL EXPERIENCE

GAOTek Inc. — Technical Intern

Aug 2025 – Nov 2025

- Engineered an automated product-documentation workflow to catalog 120+ SaaS/automation products, cutting manual entry errors by 20%+.
- Integrated Apollo, Clay, and Thunderbird APIs into a unified lead-gen + SMTP email-outreach pipeline generating 100+ qualified leads daily.

PROJECTS

Jarvis — Local-First AI Assistant with Voice

Python, FastAPI, Ollama, Voice AI

Tech Stack: Python, FastAPI, Ollama, faster-whisper, Piper TTS, openWakeWord, SQLite

- Architected a modular AI assistant OS: 7 isolated modules (brain, tools, memory, speech, wake, web, kernel) communicate only via typed async events on an in-process bus — zero inter-module imports — enabling independent replacement of any component.
- Engineered hands-free voice via wake-phrase + clap detection (crest-factor signal analysis) with local STT/TTS roundtrips — full voice I/O, zero cloud APIs, zero cost.
- Built a tool-calling agent loop with per-conversation concurrency locks, gated conversation resets (user-confirmed), and Ollama-native streaming (~3x faster on CPU).
- Validated with a 30+ test suite using mock LLM providers, HTTP transports, and fake audio for zero-dependency CI.

Wanna Watch — AI-Powered Movie Discovery Platform

React, Node, FastAPI, ML

Tech Stack: React 19, Node.js, FastAPI, Python, scikit-learn, PostgreSQL, Vercel, Render

- Designed a hybrid recommendation engine combining Content-Based Filtering (TF-IDF & Cosine Similarity) with real-time TMDb API lookups.
- Formulated a 5-question personality quiz algorithm outputting dynamic user-matching scores to drive engagement.
- Developed full-stack user services with JWT auth, stateful favorites/watchlists, and responsive lazy-loaded sections.
- Orchestrated microservices deployment on Vercel (frontend) and Render (backend + ML).

CleanFlow — Automated Data Pipeline

Python, FastAPI, ETL

Tech Stack: Python, FastAPI, Pandas, NumPy, RESTful APIs, Excel Reporting

- Engineered an ETL pipeline to ingest, validate, and standardize messy real-world datasets, saving up to 85% of manual cleaning time.
- Implemented business-rule validation for format accuracy (emails, phones, dates) and schema consistency, with structured Excel anomaly reports.

TaskForce — Full-Stack Productivity Platform

Node, Express, PostgreSQL, MySQL

Tech Stack: Node.js, Express, PostgreSQL, MySQL, Vanilla JS, REST API, JWT

- Overhauled a legacy task app into a high-performance dark-themed brutalist UI with modular layout blocks and custom CSS typography.
- Scaled an MVC Express backend, optimizing queries and connection pools across PostgreSQL and MySQL.
- Implemented JWT authentication and secure input validation for server-side security.

EDUCATION

Swami Vivekananda University — B.Tech in Computer Science & Engineering

2024 – 2027 (Expected)

- Grade: CGPA: **8.95 / 10.0**

Government Polytechnic Khutri — Diploma in Computer Engineering

2021 – 2024

- Grade: **Distinction**

CERTIFICATIONS

CS50's Introduction to AI with Python

HarvardX (2025)

CS50's Introduction to Computer Science

HarvardX (2024)

Software Engineering

NPTel / IIT Kharagpur (2024)

OSINT Using Python

CyberVidyaPeeth (2023)